Aryan Trivedi

B. Tech
Computer Science and Engineering
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Education

Indian Institute of Information Technology Ranchi *Aug 2024 – Present*
Bachelor of Technology in Computer Science and Engineering CGPA : 9.22

Technical Skills

Languages: Java, Python, Dart, TypeScript, SQL
Frameworks: Flutter, Next.js, React.js, Node.js, Express.js
Cloud/Databases: PostgreSQL, Supabase, Firebase, Vercel
AI / APIs: Google Gemini API, Serper API, NLP Pipelines
Developer Tools: Git, GitHub, Postman, Figma
Coursework: Data Structures & Algorithms, DBMS, OOP

Experience

Navicon Infraprojects Pvt. Ltd. *Sep 2025 – Present*
Software Engineering Intern India

- Developed a **cross-platform construction ERP application** using **Flutter (Dart)** for inventory, finance, and project management workflows.
- Built backend services with **Supabase and PostgreSQL**, implementing authentication, role-based access control (RBAC), and real-time data synchronization.
- Designed normalized relational database schemas for **5+ core ERP entities**, improving data consistency and reducing query response time by **30%**.

Palisadoes Foundation
Open Source Contributor Remote

- Authored **unit and integration test suites**, achieving **100% test coverage** for **2 UI components**.
- Increased repository-wide test coverage by **0.4%**, improving regression detection and release confidence.
- Identified edge cases and validated multi-path execution logic through automated testing and coverage analysis.

Personal Projects

The Archive: Rich-text based Blogging Site *Jan 2026*
Tools: TypeScript, React, Express, Postgres, Multer

- Built a **fast, production-ready publishing platform** with a polished rich-text editor and stable server APIs, using instant rebuilds and strict TypeScript checks to catch bugs early and ensure reliability.
- Designed a **compact rich-text content format and parser** that preserves poetry line breaks, supports inline images and links, and stores content as predictable JSON for safe rendering and editing.
- Implemented a **robust image upload pipeline** with type and size validation, server-side optimization (resize), versioned storage, and CDN delivery; ensured safe rollback on upload failures.
- Built **secure authentication and session management** with short-lived access tokens, rotating refresh tokens with server-side revocation, strong salted password hashing, and downloadable base64 recovery files.

Synapse – AI Knowledge Blueprinting Platform *Jun 2025*
Tools: Next.js, TypeScript, React Flow, Gemini API, Serper API

- Built a **Next.js + TypeScript platform** that transforms abstract topics into structured conceptual blueprints.
- Engineered **meta-prompting pipelines with Gemini** to generate hierarchical knowledge graphs.
- Implemented **React Flow** for interactive graph editing, supporting **100+ nodes** per blueprint without UI lag.
- Applied **depth-first traversal (DFS)** on graph structures to recursively expand nodes with controlled depth, optimizing runtime and avoiding exponential growth.
- Automated research using **Serper API and AI summarization**, reducing manual research time by **60%**.

Achievements

Competitive Programming *2024 – Present*

- Achieved rating of **1643 (Highest)** in **Codechef**.
- Solved **400+ problems** across arrays, recursion, trees, and basic dynamic programming.

Naukri Campus Engineer's Ring of Honour *2025*

- Secured **AIR 34** in Naukri Campus Engineer Ring of Honour among a large national applicant pool.
- Evaluated on **coding skills, analytical thinking, and core computer science fundamentals**, and **AI**.